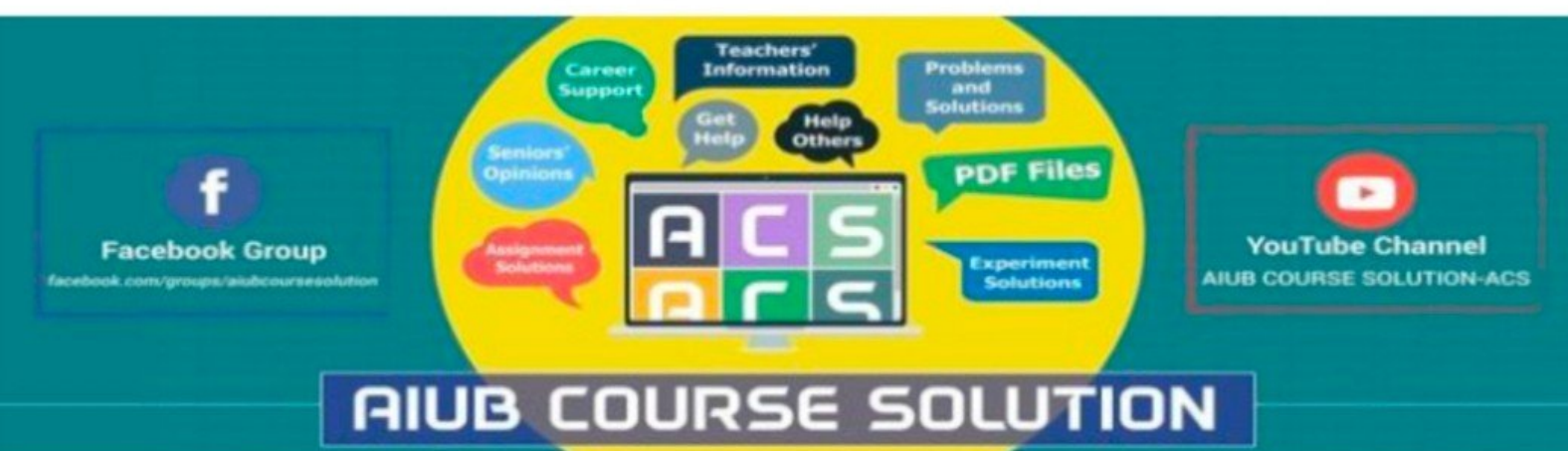




TOGETHER WE CAN ACHIVE MORE

COURSE NAME: Object Oriented Programming 1
SEMESTER: Summer 24-25 (Lecture 2)

SOLVED BY

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Constructor → Class ଏବଂ ନାମରେ ଆକରଣ ।

- Just object create କଲେ enough.
- Return Type ନାହିଁ ।

default → parameter ଏବଂ ଡିଫଲ୍ଟ କିନ୍ତୁ ଆକରଣ ନା ।
Constructor

person () ଡିଫଲ୍ଟ ଆକରଣ constructor use କରାଯିବ ବୋଲି ଥିବ ।

person (int a) ଡିଫଲ୍ଟ ଆକରଣ parameterized constructor use କରାଯିବ ବୋଲି ଥିବ ।

overload → parameter ଭିନ୍ନ ଭିନ୍ନ 2ଟି method

constructor, String is immutable → msg

Output tracing → String, array, constructor.

Object creation ଏବଂ through ଡିଫଲ୍ଟ String ~~କ~~
ଏବଂ value ଯାଞ୍ଚ କରାଯିବ ।

Slide 2.2 $\Rightarrow$ class and object

All Bold lines $\rightarrow$ For MCQ

- **Classes are templates or blueprints for object**
- **Object is an instance of class.**
- **The process of creating an object is called instantiation.**
- **Attributes of an object $\rightarrow$ instance variable.**
- **Methods of an object $\rightarrow$ instance methods**
- **In java, object are created using the new keyword**

```
Employee anEmployee = new Employee();
```

Defining A class $\rightarrow$

The keyword "class" followed by the name of the class.

Instance variables →

- variables can be prefixed with a visibility modifier.
- variables can have one of 4 different visibilities.
 - public
 - private
 - protected
 - default.

Null References →

- Null means refers to no object.
- Null is the default value of an object reference before it is initialized.

Constructors

- There is no ~~data~~ return type. Not even void.
- The method name is the same name as the class.
- Constructors can be overloaded

Slide 3.1 $\Rightarrow$

Variable types $\rightarrow$ variables can be divided into 3 types.

- Local variables.
- Class variables
- Instance variables

Bold lines $\rightarrow$

Class variable $\rightarrow$

- Used static keyword.
- We do not need any object to access it. It is accessed using the class name.

Instance variables $\rightarrow$ not used Static keyword.

- We have to use an object to access it.

~~Variable~~ $\rightarrow$

27-07-25

Slide 3.2 =>

❑ A static block is a normal block of code enclosed in braces, {}, and preceded by the static keyword.

❑ A static block is also called static initialization block.

❑ A static block executed only once, when a program begins execution.

Slide 3.3 =>

❑ Strings, which are widely used in Java programming, are a sequence of characters.

❑ In Java, Strings are object

❑ String is the only class which has implicit instantiation.

❑ The String class is defined in the java.lang package.

☐ Strings are immutable, ~~***~~

* Scanner class

- It is found in the java.util package.
- Used to get user input.
- By the help of scanner in java, we can get input from the user in primitive types such as int, long, double, byte, float, Short, etc

29-07-25.

MCQ $\rightarrow$ Encapsulation.

extend class এর মাধ্যমে টাকে inherit করা হয়ে থাকে তাই এটি parent class.

1টা parent class and 1টা child class
আরও single class.

1টা parent class আরও and 2টা
child class আরও Hierarchical.

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